KinoPlex: sequence meets structure

An AI-enabled structural atlas decodes kinase specificity across the human proteome

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Which plausible motifs are physically usable?

About 1.8 million human serine, threonine and tyrosine (STY) residues could accept phosphate.

- Only about 6% have experimental phosphorylation evidence.
- Only about 5% of those have an assigned upstream kinase.
- A peptide motif measures catalytic sequence preference; the same sequence in a folded protein may be buried or geometrically inaccessible.

Central question
Can a proteome-wide structural prior distinguish plausible phosphorylation sites and make motif-based kinase assignments more useful?

Article abstract; Supplementary Note II, motivation.

Two distinct questions, two complementary scores

Structural phospho-competency

Input: the local environment of an STY in a predicted full-length structure.

Output: a site-level score for structural permissiveness.

It does not specify the kinase or guarantee phosphorylation in a particular cell.

Kinase motif compatibility

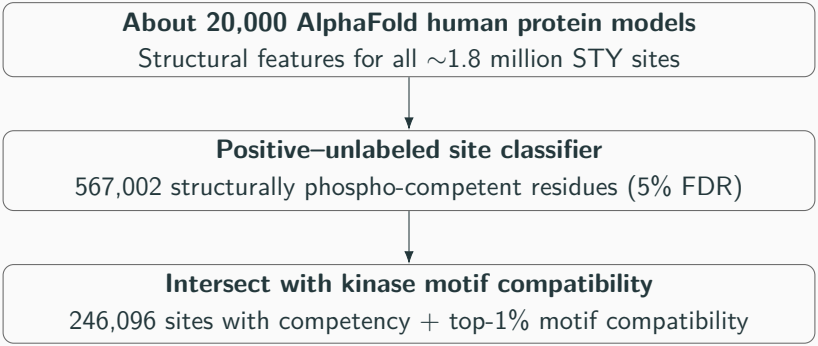
Input: the amino-acid sequence around the STY and a kinase position-specific scoring matrix (PSSM).

Output: motif score percentile relative to the whole proteome.

It does not ensure that the motif can be reached in the folded protein.

Supplementary Note III, Methods 1–2; Extended Data Fig. 2.

The atlas is an intersection of sequence and structure



Candidate sites are predictions of potential, not 250,000 newly observed phosphorylation events.

Fig. 1f, Fig. 5a; Results. Intersection: 164,069 S + 70,088 T + 11,939 Y, at 5% competency FDR.

Represent the chemical environment of the acceptor

Feature family	What it asks about the STY residue
Confidence and flexibility	pLDDT, low-confidence runs, order–disorder boundaries
Accessibility	Total and hydroxyl-specific solvent-accessible surface area ($Å^2$)
Local crowding	Packing, neighbor counts, exposure, cavity openness and depth
Geometry	Backbone torsions, side-chain angles, secondary-structure context
Microenvironment	Charge, polarity, hydrophobicity and approximate hydrogen-bond contacts

More than 100 descriptors combine local atomic coordinates and sequence-window features. These include heuristic proxies, not a physical simulation of kinase binding.

Supplementary Methods, sections 1–4.

Learning from known positives and unknown sites

Labels: about 116,000 curated sites are positive; remaining STYs are unlabeled, not established negatives.

1. Fit separate models for serine, threonine and tyrosine; hold out 20% of sites, retaining label proportions.
2. Use XGBoost (depth 4, learning rate 0.1, threefold cross-validation) to discriminate known from unlabeled sites.
3. Estimate the hidden-positive fraction to correct evaluation: unlabeled residues contain real, undiscovered phosphosites.

Why the distinction matters
A classifier's uncorrected AUC against unlabeled sites and its positive–unlabeled (PU) corrected AUC answer different evaluation questions.

Fig. 1d–f; Methods, PUC transfer learning; Supplementary Note I, sections 2–4.

PU correction makes a strong sampling assumption

Let $y = 1$ mean truly phosphorylable and $s = 1$ mean a labeled known site. The model initially estimates $g(x) = P(s = 1 \mid x)$.

$$P(s = 1 \mid y = 1, x) = c \quad \Longrightarrow \quad P(y = 1 \mid x) = g(x)/c.$$

Assumption: labeled positives are sampled completely at random from the true positive population (SCAR).

After isotonic calibration, estimate c using held-out labeled positives and infer hidden-positive prevalence in the unlabeled pool.

Main-model evidence: raw AUCs S/T/Y = 0.84/0.84/0.83; PU-corrected AUCs = 0.92/0.92/0.91. These differ from the serine ablation benchmark on the next slides.

Fig. 1e; Methods, Elkan–Noto class prior estimation. Assumption may be challenged by discovery bias.

A running example: a good motif can be an unusable site

Candidate	Sequence match	Structural presentation
A	High PSSM percentile	Buried hydroxyl, dense local packing
B	High PSSM percentile	Exposed flexible segment
C	Modest PSSM percentile	Exposed surface of an ordered domain

- Motif-only ranking favors A and B; competency should suppress A.
- Disorder alone may miss C: ordered does not mean inaccessible.
- C may remain biologically relevant even below an extreme motif cutoff.

Illustrative schematic, not experimental sites. Logic: Supplementary Notes I–II.

Does the classifier add more than disorder and exposure?

Same held-out serine split

Model	Corrected AUC
Disorder logistic	0.822
SASA logistic	0.730
Disorder + SASA	0.821
Full XGBoost	0.950

Hold learner class fixed

Raw AUC for XGBoost feature ablations:

- Two-feature model: 0.75
- Orthogonal feature group: 0.84
- Full model: 0.87

Interpretation: the gain is not solely a change from logistic regression to boosted trees. Keep corrected and raw AUCs separate.

Supplementary Note I, section 8.1; Supplementary Figs. 1–2.

The added value includes structured phosphosites

At balanced thresholds, comparison with disorder-only prediction on known phosphoserines gives:

12,436 sites (16.1%)
Recovered by full KinoPlex but missed by the disorder threshold; span 5,847 proteins.

2,338 sites (3.0%)
Recovered by disorder but missed by the full model.

Pyruvate kinase (PKM) is a structural vignette: exposed regulatory sites on folded domains can be competent despite low disorder.

Qualification: this rescue count summarizes known sites; it is not itself an independent prospective test of new sites.

Supplementary Note I, sections 8.2–8.3; Supplementary Figs. 1d, 3–4.

Combined scores recover known kinase substrates better

Benchmark: 10,540 PhosphoSitePlus kinase–substrate sites across 166 kinases, ranked against proteome-wide STY background.

Ranking paradigm	Global ROC-AUC	Recall at 5% FPR
Motif PSSM only	0.837	53%
Motif + structural competency	0.951	78%

Improved rankings also appear in examples including CDC7, MTOR, GSK3B and CK1A.

Evaluation limit: this curated-substrate benchmark supports prioritization; it does not constitute a held-out kinase-perturbation experiment. Overlap with site-level competency labels may favor well-studied sites.

Fig. 3a–b and Results. FPR is false-positive rate against background, distinct from site-call FDR.

Sequence–structure coupling changes effective specificity

For each kinase, score sites proteome-wide and inspect the structural competency of its top 1% motif hits.

Positive-selecting kinases

Preferred motif instances often occur in exposed, flexible environments.

Example: ERK1-associated PXXSP motifs skew toward disordered, accessible contexts.

Negative-selecting kinases

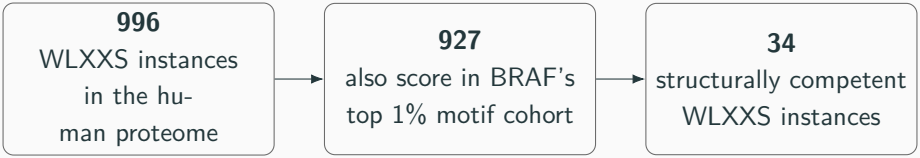
Preferred motif instances often occur in ordered or occluded environments.

Example: BRAF-associated WLXXS motifs skew toward high confidence, low accessibility and dense packing.

These are population-level associations between motif composition and structural presentation, not a demonstrated evolutionary selection mechanism.

Extended Data Figs. 4, 7–8; Supplementary Note II.

BRAF illustrates a severe structural bottleneck



WLXXS motifs have median pLDDT around 90 and low accessibility; ERK1's PXSP motif has median pLDDT around 40.

Interpretation: many excellent peptide motifs are unavailable in intact proteins. The surviving sites still need experimental kinase assignment.

Fig. 4d–e; Results, motif-level structural constraints. Counts concern one illustrative motif.

Compartment context: FAM20C in secreted proteins

- Test 41,436 S/T sites across 694 secreted proteins against nonsecreted proteins.
- Motif score distributions do not differ significantly between compartments.
- At stronger FAM20C motifs, structural competency rises more sharply in secreted proteins.
- The localization-specific pattern extends to orthologs across six species; conserved $+0$ and $+2$ residues fit the SxE motif.

What this adds

Sequence–structure coupling depends on a kinase's accessible protein compartment.

Evolutionary conservation supports the association; it does not establish how selection produced it.

Fig. 5c–d; Extended Data Fig. 9; Results, compartment-specific conservation.

An independent substrate reference supports the coupling

Compare peptide-array PSSMs with sequence motifs derived from curated cellular substrates; include kinases with more than 20 annotated sites.

- Motif agreement varies: JNK2 $r \approx 0.88$, PKACA $r \approx 0.87$; ULK1 $r \approx 0.05$, LRRK2 $r \approx 0.18$.
- Stronger negative selectivity associates with weaker motif agreement: Spearman $\rho \approx -0.629$.
- Negative selectivity also associates with lower PSSM percentiles among known cellular substrates: $\rho \approx -0.556$.

Practical implication
An extreme peptide-motif cutoff can discard real substrates of negative-selecting kinases. Structural filtering complements motif scores; it does not make motif scoring universally sufficient.

Supplementary Note II, sections 2–9; Supplementary Figs. 7–10.

Infer kinase activity from which sites were detected

Define a candidate substrate pool for kinase k from high motif scores plus structural competency. Compare detected substrates with a background expectation.

$$\text{enrichment}_k \sim \frac{\text{observed detected candidate substrates of } k}{\text{expected detections given its plausible substrate pool}}$$

- External validation uses PSSM >99th percentile and competency at 1% FDR, then standardizes activity across cell lines.
- The input is site identification, without requiring relative phosphopeptide intensities.
- A PSSM-only background assigns similarly sized candidate pools to kinases; structure changes the kinase-specific expected opportunity.

Schematic relationship; observed/expected enrichment rather than an exact score formula. Supplementary Note III, Methods 1–2.

New site discovery in K562 cells

Experiment: 12 h serum/nutrient deprivation, 30 min serum re-stimulation; IMAC enrichment, 24 fractions and Orbitrap Astral DIA phosphoproteomics.

Evidence layer	Sites
Identified at <1% FDR	40,908
Localization probability >0.5	35,892
Mapped to canonical isoforms	30,410
Mapped and predicted competent	27,055 (89%)
Novel versus PhosphoSitePlus	6,327
Novel and predicted competent	4,844 (76.6%)

Fig. 6a–b; Results; Methods, cell culture and MS. Novel-site recovery supports generalization.

Identification-only activity recovers K562 biology

- ABL is the most enriched tyrosine kinase: over 5% of its high-motif, competent candidate sites are detected.
- K562 carries constitutively active BCR–ABL; 12 of 15 detected known ABL substrates are predicted competent.
- HER2 shows the lowest inferred tyrosine-kinase activity, consistent with absent ERBB2 expression.
- TAM-family kinases TYRO3, AXL and MERTK are also enriched.

Control logic: agreement with established cell-line biology supports the enrichment score. A single stimulated endpoint does not quantify a stimulus-induced activity change.

Fig. 6c; Results, kinase enrichment capacities. Activity candidate pool uses 1% competency FDR.

External functional evidence: inhibitor sensitivity

125 cancer cell lines, over 55,000 phosphosites and over 6,000 compounds connect inferred kinase activities to drug response.

Activity / inhibitor class	AUC	BH-adjusted p
MEK1 / MEK inhibitors	0.81	9.9×10^{-8}
MEK2 / MEK inhibitors	0.79	5.3×10^{-7}
MTOR / MTOR inhibitors	0.76	6.2×10^{-8}

These AUCs describe enrichment of cognate inhibitor classes in ranked drug associations, not patient response prediction.

PSSM-only scoring degraded class enrichment (MEK1 AUC 0.69). Drug polypharmacology and lineage can still confound individual associations.

Supplementary Note III, sections 2, 5.2; Supplementary Figs. 13–15. Adjusted p: note text.

External site-level evidence: reproducibility across tumors

CPTAC: 3,205 samples, 10 cancer types, two acquisition centers.

- Competency recovery rises from about 79% for rarely detected sites to over 93% for sites seen in more than 90% of samples.
- The trend persists across cancer types and centers.
- Activity profiles group by cancer type, with coherent cancer-specific programs across centers.

Evidence stratification: PhosphoSitePlus recovery is about 55% for single-evidence sites and above 90% with at least 10 observations.

Caution: detection frequency combines biological prevalence, protein abundance and peptide detectability. Weak support does not prove an annotation is false.

Fig. 6g–h; Supplementary Note III, sections 3, 6–8; Supplementary Figs. 17–23.

What remains unresolved?

1. **Static structure:** AlphaFold confidence is an imperfect disorder proxy; conformational changes, complexes and modifications alter accessibility.
2. **Learning assumptions:** positives are not a random sample of real sites. PU calibration depends on labeling assumptions; random site splits do not establish transfer to unseen proteins.
3. **Causal mapping:** a kinase-compatible, competent site is still a candidate. Localization, docking, abundance and phosphatases influence cellular phosphorylation.
4. **Activity interpretation:** identification enrichment is a functional proxy; drug correlations do not prove a direct kinase–site relationship or clinical utility.

Our assessment informed by Supplementary Notes I–III and feature-extraction methods.

Discussion and source scope

Take-home interpretation
Structural opportunity is a useful prior for motif-based signaling inference. The strongest evidence is added value beyond simple disorder baselines and independent links to reproducible sites and inhibitor classes.

Discussion: Would a protein-held-out benchmark, perturbation-resolved kinase knockouts or dynamic ensemble structures most change our confidence?

Resources: interactive atlas: kinoplex.phosphosite.org; structural-feature code: github.com/dvanderwall/KinoPlex; data: doi:10.5281/zenodo.20633591.

Journal main text, Methods, extended data and the 73-page supplement were reviewed. Quantitative panels are identified on each slide.

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